

Adding, Subtracting, and Multiplying Polynomials

Answer Key

Algebra 2

Quick Answers

1. $4x^3 - 2x^2 + 2$

2. $7x^4 + 2x^2$

3. $3x^3 - 7x + 12$

4. $x^3 + x^2 - x + 15$

5. $3x^4 + 4x^3 - 5x + 8$

6. 25

7. $2x^4 + 8x^3 - 5x^2 - 12x + 3$

8. $(x - 2)(x + 1)(x + 2)$

9. $x^4 + 2x^3 - x^2 - 2x + 1$

10. $x^6 - 4x^3 + 4$

Common wrong answers

6. *If they got 10: multiplied only the x through the trinomial and never distributed the 3*

How to use this key. Each solution shows the model filled in, then the collected polynomial. Watch for the two errors this sheet is built around: combining terms that are not like terms, and distributing a minus sign or a factor to only the first term inside a parenthesis.

1. Add $(3x^3 + 2x^2 - 5) + (x^3 - 4x^2 + 7)$.

Solution: Line the terms up by power. The x^3 column holds $3x^3 + x^3 = 4x^3$; the x^2 column holds $2x^2 + (-4x^2) = -2x^2$; the x column is empty in both polynomials; the constant column holds $-5 + 7 = 2$. Only coefficients are added — the exponents never change, because adding like terms is the distributive property run backwards: $3x^3 + x^3 = (3 + 1)x^3$. Collected in descending order, $\boxed{4x^3 - 2x^2 + 2}$.

2. Add $(2x^4 - x^2 + 6x) + (5x^4 + 3x^2 - 6x)$.

Solution: x^4 : $2x^4 + 5x^4 = 7x^4$. x^2 : $-x^2 + 3x^2 = 2x^2$. x : $6x + (-6x) = 0$, so that column vanishes entirely and no x term appears in the answer. Losing a column lowers no degree here — the leading x^4 column survived, so the sum still has degree 4. The result is $\boxed{7x^4 + 2x^2}$.

3. Subtract $(4x^3 - 2x + 9) - (x^3 + 5x - 3)$.

Solution: The subtraction sign belongs to the whole second polynomial, so distribute it first: $-(x^3 + 5x - 3) = -x^3 - 5x + 3$. Note the last term became $+3$. Now add columns: $4x^3 - x^3 = 3x^3$; $-2x - 5x = -7x$; $9 + 3 = 12$. The difference is $\boxed{3x^3 - 7x + 12}$.

4. Multiply $(x + 3)(x^2 - 2x + 5)$.

Solution: The area box has two rows and three columns, so six products.

$\times$	x^2	$-2x$	5
x	x^3	$-2x^2$	$5x$
3	$3x^2$	$-6x$	15

Adding the cells and collecting like terms: x^3 stands alone; $-2x^2 + 3x^2 = x^2$; $5x - 6x = -x$; the constant is 15. So the product is $x^3 + x^2 - x + 15$.

5. Subtract $(5x^4 + 3x^3 - x) - (2x^4 - x^3 + 4x - 8)$.

Solution: Distribute the minus to all four terms: $-2x^4 + x^3 - 4x + 8$. The two already-negative terms flip to positive, which is the step this problem tests. Adding columns: $5x^4 - 2x^4 = 3x^4$; $3x^3 + x^3 = 4x^3$; $-x - 4x = -5x$; the constant is $0 + 8 = 8$. The difference is $3x^4 + 4x^3 - 5x + 8$.

6. Check $(x + 3)(x^2 - 2x + 5) = x^3 + x^2 - x + 15$ at $x = 2$.

Solution: Factored form: $x + 3$ is 5, and $x^2 - 2x + 5$ is $4 - 4 + 5 = 5$, so the product is $5 \cdot 5 = 25$. Expanded form: $8 + 4 - 2 + 15 = 25$. The two agree, which is strong evidence the expansion is right — a single value cannot prove it, but any disagreement would disprove it immediately. Had the 3 never been distributed, the expansion would have been $x^3 - 2x^2 + 5x$, worth only 10 at $x = 2$, and the check would have caught it.

7. Multiply $(2x^2 - 3)(x^2 + 4x - 1)$.

Solution: Six cells again.

$\times$	x^2	$4x$	-1
$2x^2$	$2x^4$	$8x^3$	$-2x^2$
-3	$-3x^2$	$-12x$	3

The only like pair is $-2x^2$ and $-3x^2$, which combine to $-5x^2$; everything else stands alone. Collected:

$$2x^4 + 8x^3 - 5x^2 - 12x + 3$$

8. Factor $x^3 + x^2 - 4x - 4$ by grouping.

Solution: Group as $(x^3 + x^2) + (-4x - 4)$. The first group gives $x^2(x + 1)$; the second gives $-4(x + 1)$. The common factor $(x + 1)$ comes out: $(x + 1)(x^2 - 4)$. The second factor is a difference of squares, so it splits again: $(x - 2)(x + 2)(x + 1)$. This matches the claimed factors, so the expansion was correct.

9. Expand $(x^2 + x - 1)^2$.

Solution: A square is a product of the trinomial with itself, so the box is three by three — nine cells, not three.

$\times$	x^2	x	-1
x^2	x^4	x^3	$-x^2$
x	x^3	x^2	$-x$
-1	$-x^2$	$-x$	1

Collecting: x^4 ; $x^3 + x^3 = 2x^3$; $-x^2 + x^2 - x^2 = -x^2$; $-x - x = -2x$; constant 1. So

$$\boxed{x^4 + 2x^3 - x^2 - 2x + 1}$$

10. A student writes $(x^3 - 2)^2 = x^6 - 4$.

Solution: The student squared each term separately, treating “squaring” as if it distributed over subtraction. It does not: squaring is multiplication by the whole binomial, so the box has four cells, and the two cross terms are exactly what was dropped.

$\times$	x^3	-2
x^3	x^6	$-2x^3$
-2	$-2x^3$	4

The two cross cells sum to $-4x^3$, and the last cell is $+4$, not -4 , because a negative times a negative is positive. The correct expansion is $\boxed{x^6 - 4x^3 + 4}$.